22. In the electrolytic refining of copper, which one of the following is used as anode?

a) Pure copper
b) Impure copper
c) Carbon rod
d) Platinum electrode

23. Which of the following plot gives Ellingham diagram

a) ΔS Vs T
b) ΔG^0 Vs T
c) ΔG^0 Vs $\frac{1}{T}$
d) ΔG^0 Vs T^2

24. In the Ellingham diagram, for the formation of carbon monoxide

a) $\left(\frac{\Delta S^0}{\Delta T}\right)$ is negative
b) $\left(\frac{\Delta G^0}{\Delta T}\right)$ is positive
c) $\left(\frac{\Delta G^0}{\Delta T}\right)$ is negative
d) initially $\left(\frac{\Delta T}{\Delta G^0}\right)$ is positive, after 700^0C, $\left(\frac{\Delta G^0}{\Delta T}\right)$ is negative

25. Which of the following reduction is not thermodynamically feasible?

a) $Cr_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Cr$
b) $Al_2O_3 + 2Cr \longrightarrow Cr_2O_3 + 2Al$
c) $3TiO_2 + 4Al \longrightarrow 2\,Al_2O_3 + 3Ti$
d) none of these

26. Which of the following is not true with respect to Ellingham diagram?

a) Free energy changes follow a straight line. Deviation occurs when there is a phase change.
b) The graph for the formation of CO_2 is a straight line almost parallel to free energy axis.
c) Negative slope of CO shows that it becomes more stable with increase in temperature.
d) Positive slope of metal oxides shows that their stabilities decrease with increase in temperature.

Answer the following questions:

1. What is the difference between minerals and ores?
2. What are the various steps involved in extraction of pure metals from their ores?
3. What is the role of Limestone in the extraction of Iron from its oxide Fe_2O_3?
4. Which type of ores can be concentrated by froth floatation method? Give two examples for such ores.
5. Out of coke and CO, which is better reducing agent for the reduction of ZnO? Why?
6. Describe a method for refining nickel.
7. Explain zone refining process with an example using the Ellingham diagram given below.

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